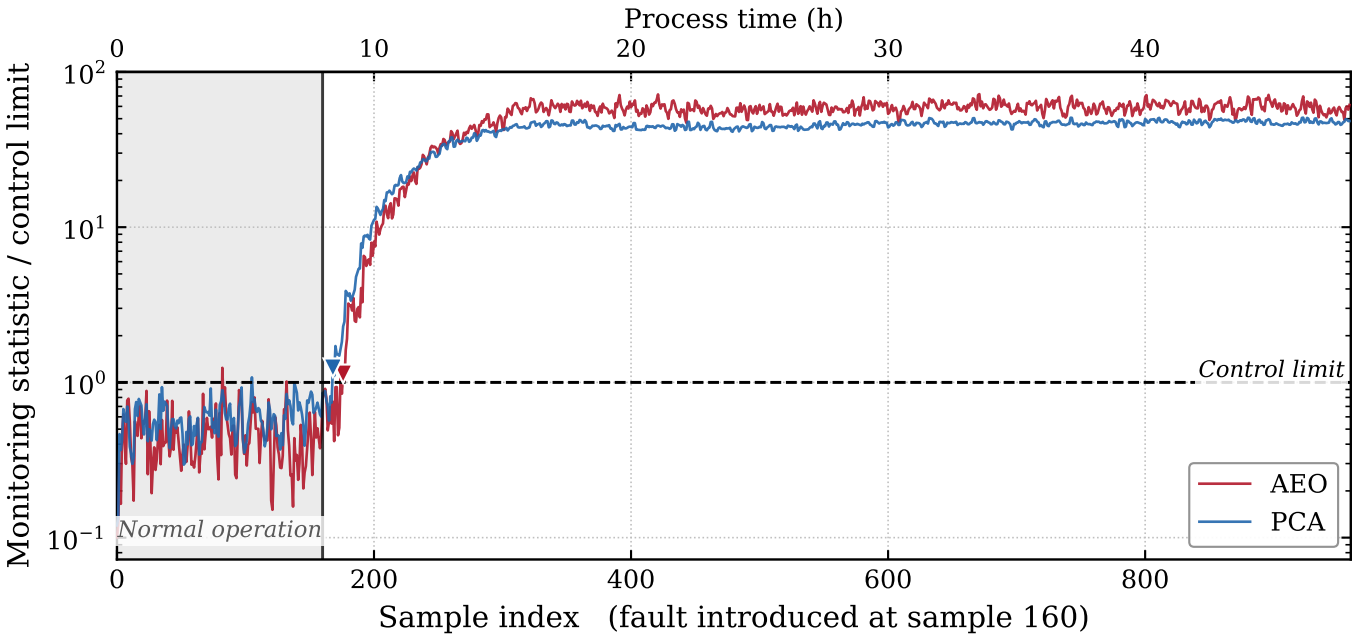
Fault Detection Performance for IDV(2)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	19.0 samples (57 min)	MDR 0.022	detected in 7/7 runs
PCA	delay	9.7 samples (29 min)	MDR 0.012	detected in 7/7 runs

PCA detects 9.3 samples (28 min) earlier; PCA missed-detection rate lower by 44%